

Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introduction to Computer Architecture
Homework 9 (Due 11/24)

1. Consider the following MIPS assembly program running on a single-issue **R10K-like processor**, as discussed in class (F, De, Di, S, X, C, R). A snippet of the corresponding pipeline diagram is shown below:

```

add  $t0, $t0, $s0
lw   $t1, 0($t1)
and  $t2, $t0, $s1
slt  $t2, $t2, $s2
beq  $t2, $zero, SKIP
lw   $t1, 0($t0)
SKIP: sw  $t1, 0($t0)
      sub $t2, $t1, $s2
      or  $s2, $t2, $t0
      add $s1, $t0, $s1
      lw  $s2, 0($s0)

```

insn\cycle	1	2	3	4	5	6	7	8	9	10	11	12	13
add \$t0, \$t0, \$s0	F	De	Di	S	X	C	R						
lw \$t1, 0(\$t1)		F	De	Di	S	X	C	R					
and \$t2, \$t0, \$s1			F	De	Di	S	X	C	R				
slt \$t2, \$t2, \$s2				F	De	Di	S	S	X	C	R		
beq \$t2, \$zero, L1					F	De	Di	S	S	S	X	C	R
lw \$t1, 0(\$t0)						F	De	Di	S	X	C	=	
sw \$t1, 0(\$t0)							F	De	Di	S	S	=	
sub \$t2, \$t1, \$s2								F	De	Di	S	=	
or \$s2, \$t2, \$t0									F	De	Di	=	
add \$s1, \$t0, \$s1										F	De	=	
lw \$s2, 0(\$s0)											F	=	
sw \$t1, 0(\$t0)													F

Assume that all instructions take 1 cycle in the execute stage on this processor (including branches and stores). When running the program above, the outcome of the branch instruction (i.e., beq \$t2, \$zero, L1) is computed on cycle 11 and is found to be mispredicted (assume it was predicted to be not taken). This means that on the next cycle, all instructions fetched after the branch must be flushed. The contents of the reservation stations, map table, free list and reorder buffer **just before flushing the pipeline (i.e., at the end of cycle 11)** are shown below for this processor. Registers are popped from the left of the free list and pushed onto the right of the free list. And in the reorder buffer, instructions are listed from oldest (top) to youngest (bottom).

Reservation Stations and Functional Units					
Functional Unit	Instruction	Operand (rs)		Operand (rt)	
		Physical Reg	Ready (Y/N)	Physical Reg	Ready (Y/N)
add/sub/slt					
and/or	or \$s2, \$t2, \$t0	P5	N	P12	Y
beq					
lw					
sw					

Map Table		
Architectural Reg	Physical Reg	Ready (Y/N)
\$t0	P12	Y
\$t1	P6	Y
\$t2	P5	N
\$s0	P4	Y
\$s1	P9	Y
\$s2	P11	N

Free List	P2, P10, P1
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Reorder Buffer		
Instruction	Dest Physical Reg (rd)	Old Physical Reg (rd)
beq \$t2, \$zero, L1		
lw \$t1, 0(\$t0)	P6	P3
sw \$t1, 0(\$t0)		
sub \$t2, \$t1, \$s2	P5	P7
or \$s2, \$t2, \$t0	P11	P8

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- (a) Assume that flushing the processor pipeline upon a branch misprediction only takes 1 cycle, as shown in the pipeline diagram above. Fill in the contents below of the reservation stations, map table, free list and reorder buffer **immediately after flushing the pipeline (i.e., at the end of cycle 12)**.

Answer:

Reservation Stations and Functional Units					
Functional Unit	Instruction	Operand (rs)		Operand (rt)	
		Physical Reg	Ready (Y/N)	Physical Reg	Ready (Y/N)
add/sub/slt					
and/or					
beq					
lw					
sw					

Map Table		
Architectural Reg	Physical Reg	Ready (Y/N)
\$t0	P12	Y
\$t1	P3	Y
\$t2	P7	Y
\$s0	P4	Y
\$s1	P9	Y
\$s2	P8	Y

Free List	P6, P5, P11, P2, P10, P1
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Reorder Buffer		
Instruction	Dest Physical Reg (rd)	Old Physical Reg (rd)
beq \$t2, \$zero, L1		

- (b) After flushing the pipeline, the processor begins fetching from the correct (branch-taken) path on cycle 13. The pipeline diagram is shown below for the remaining instructions in the program.

insn\cycle	13	14	15	16	17	18	19	20	21	22	23	24	25	26
sw \$t1, 0(\$t0)	F	De	Di	S	X	C	R							
sub \$t2, \$t1, \$s2		F	De	Di	S	X	C	R						
or \$s2, \$t2, \$t0			F	De	Di	S	S	X	C	R				
add \$s1, \$t0, \$s1				F	De	Di	S	X	C	R	R			
lw \$s2, 0(\$s0)					F	De	Di	S	X	C	R	R		

Fill in the contents below of the reservation stations, map table, free list and reorder buffer **at the end of cycle 18**.

Answer:

Reservation Stations and Functional Units					
Functional Unit	Instruction	Operand (rs)		Operand (rt)	
		Physical Reg	Ready (Y/N)	Physical Reg	Ready (Y/N)
add/sub/slt	add \$s1, \$t0, \$s1	P12	Y	P9	Y
and/or	or \$s2, \$t2, \$t0	P6	N	P12	Y
beq					
lw					
sw					

Map Table		
Architectural Reg	Physical Reg	Ready (Y/N)
\$t0	P12	Y

\$t1	P3	Y
\$t2	P6	N
\$s0	P4	Y
\$s1	P11	N
\$s2	P5	N

Free List	P2, P10, P1
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Reorder Buffer		
Instruction	Dest Physical Reg (rd)	Old Physical Reg (rd)
sw \$t1, 0(\$t0)		
sub \$t2, \$t1, \$s2	P6	P7
or \$s2, \$t2, \$t0	P5	P8
add \$s1, \$t0, \$s1	P11	P9